

HELIOLITE · CALCULATION

HelioLite sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-26	Amish Chadha	Budget top-up approved by Amish: R15 target \$455, script re-run
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (optics, hourly energy at reference sites, stiffness, pointing and calibration, wind and drives, stow reserve, mass, power, cost)

HelioLite sizing calculations

On paper, HelioLite meets twelve of its fifteen requirements and has two at risk; one cannot be verified at TRL 3. This version applies the decisions Amish took on 2026-09-25 (HLT-DDR-002): a stow stop and latch that carries the stowed hinge moment, preload springs on both drives, R13 relaxed to 13 kg, R7 and R10 rewired, and four calibration points over about 4 h as the default. The optics work: at the design point the mirror delivers about 211 W to the outside of the glazing and 159 W (about 15,000 lm) into the room, and a reference site at 45° N receives 0.50 kWh through the glazing on the winter solstice and 1.0 kWh on 1 February. Stowed wind survival (R9) is now met on paper, because the latch rather than the gearbox carries the stowed moment, with a factor of 2 on the assumed coefficient. The two added lines take the BOM to \$451; in version 0.3 the budget is \$455, a top-up Amish approved on 2026-09-26 (HLT-DDR-002 v0.2), so cost (R15) is met with \$4 to spare. Pointing (R4) stays at risk at the 95th percentile of calibration, and daily energy (R3) depends on siting. Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [D5], is the line of that script's output that carries it.

Safety: These calculations concern a reflected beam of nearly one sun, a glass mirror 2.2 m above the ground, worm drives with high output torque, and a mast in storm winds. They are first-principles estimates for a paper proof of concept and are not a substitute for datasheets, a structural review or test. See HLT-PRC-001, Safety.

■ Scope and method

The note checks every requirement in HLT-REQ-001 v0.4 against the design in HLT-PRC-001 v0.4 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS` and `derived()` values, so mast, yoke, mirror and clearance dimensions are the ones in the STEP files and in drawing HLT-DWG-001. It reads the BOM total from `bom/bom.csv`. Run it from the repo root with `python docs/04-calcs/sizing.py` (about 20 s, most of it the calibration simulation).

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Design point	Clear winter day at 45° N, DNI 800 W/m ² , 10 m to the target, 32° incidence on the mirror	HLT-REQ-001
Optics	Reflectance 0.93 clean, soiling factor 0.95; 0.98 allowance for mirror edges, sun-shape tails and spill; double glazing 0.75 at normal incidence, falling with angle by Fresnel reflection at four air-glass surfaces ($n = 1.52$) and path length; direct sunlight 95 lm/W; utilization factor 0.4 in a 3 x 4 m room	Engineering judgment and textbook optics
Sun and sky	Declination by Cooper's formula, solar time (no equation of time); clear-sky DNI by the Meinel model with Kasten-Young air mass	Textbook models, adequate for daily energy
Reference sites	Window 1.0 x 1.2 m, center 1.5 m high, in the north wall of a house 10 m wide, 8 m deep and 5.5 m high; mirror center 2.2 m high at site A (6 m east, 8 m north of the window), B (8 m east, 5 m north) or C (15 m due north)	Illustrative siting, not a surveyed site
Wind	Air 1.225 kg/m ³ ; face-on force coefficient 1.2; operating peak hinge moment coefficient 0.25 (normalized by $q \cdot A \cdot c$), in the range reported by Peterka and Derickson (SAND92-7009); stowed normal force coefficient 0.6 and stowed hinge moment coefficient 0.15, raised for a small chord because peak stow coefficients roughly double as chord halves (Emes et al.); yaw moment coefficient 0.1; mast drag coefficient 1.2	Rough; the stowed values are the largest uncertainty in R9
Stow latch	Stowed moment carried by a steel lug trapped between a 3 mm polyurethane (90A) stop pad and a sprung steel latch pawl at 55 mm from the elevation axis; design factor 2 on the stowed moment; polyurethane $E = 50$ MPa (assumed); stop, pawl, bracket and solenoid 0.18 kg and each spiral preload spring 0.08 kg (assumed)	HLT-DDR-002 N1, N2; engineering judgment
Drives	NMRV030-class 50:1 worm gearbox: rated output 17 N·m and backlash 1° as listed by StepperOnline ; maximum 22 N·m as listed for a Motovario 50:1 unit (RS); mass about 1.2 kg (typical of the frame size, to confirm); worm efficiency 0.4; NEMA17 48 mm motor 0.36 kg	Supplier listings; mass assumed
Materials	Steel $E = 200$ GPa, $G = 80$ GPa, pipe yield 240 MPa (ASTM A53 grade B); printed ASA $E = 1.8$ GPa, 1.07 g/cm ³ at 60 % infill; aluminum composite panel 5.5 kg/m ² ; glass 2,500 kg/m ³	Typical values
Calibration	Mount model with five unknowns: azimuth and elevation zero offsets, two base tilts and mirror cant on the torque tube. User jogs the spot to the target within 0.05° per axis (normal) at each point. True errors drawn at random: azimuth $\pm 10^\circ$, elevation $\pm 3^\circ$, tilts and cant with 1° and 0.3° standard deviation. 120 trials per case	Monte Carlo simulation
Power	Idle 0.4 W (ESP32 light sleep, RTC, drivers off); one motor at 5 W for 1 s per 30 s update; stow at 3.8 W (two coils at 1.0 A in 1.5 Ω , driver 0.3 W, ESP32 0.5 W)	Typical module data

■ A. Optics at the design point (R1, R2, R8)

The mirror intercepts $800 \text{ W/m}^2 \times 0.36 \text{ m}^2 \times 0.848 = 244 \text{ W}$. After reflectance and soiling 216 W leave the mirror, 211 W arrive at the outside of the glazing and 159 W enter the room [A2]. That is about 15,070 lm and a mean added illuminance of about 500 lx over 12 m² [A3]. R1 (200 W) and R2 (300 lx) are met. The beam carries 0.88 to 0.93 of the DNI; a flat mirror does not concentrate [A4].

The spot is the projected mirror outline plus the sun's 0.53° angular diameter. At 10 m it is about 0.65 m square, and a 0.5° beam error moves it 87 mm, so spot plus error on both sides is 0.82 m and fits a 1.0 m wide window. At 20 m the figure is 1.09 m and the beam spills [A5]. The largest distance at which spot plus 0.5° error fits a 1.0 m wide window is 16.8 m [A6]. R8 (3 to 20 m) is met mechanically; above about 17 m a wider target is needed.

■ B. Daily energy at reference sites (R3)

The TRL 2 figure used a generic day (6 h, mean DNI 650 W/m², mean cosine 0.80) and gave 0.97 kWh at the window and 0.73 kWh through the glazing [B1]. The hourly model below replaces it with sun geometry, clear-sky DNI, shading by the house and the angle at which the beam strikes the glazing.

Table 2. Clear-day energy through the glazing at three illustrative sites, 45° N [B2].

Site	Mirror position (east, north of window)	Distance	Glazing incidence (τ)	Day	Sunlit tracking	Mean cosine	At window	Into room	Peak into room
A	6 m, 8 m	10.0 m	37° (0.73)	21 Dec	4.8 h	0.83	0.69 kWh	0.50 kWh	152 W
A	6 m, 8 m	10.0 m	37° (0.73)	1 Feb	8.3 h	0.86	1.38 kWh	1.00 kWh	166 W
B	8 m, 5 m	9.5 m	58° (0.61)	21 Dec	5.5 h	0.75	0.66 kWh	0.40 kWh	97 W
B	8 m, 5 m	9.5 m	58° (0.61)	1 Feb	6.2 h	0.73	0.82 kWh	0.50 kWh	107 W
C	0 m, 15 m	15.0 m	3° (0.75)	21 Dec	8.4 h	0.95	1.34 kWh	1.00 kWh	163 W
C	0 m, 15 m	15.0 m	3° (0.75)	1 Feb	9.4 h	0.93	1.64 kWh	1.23 kWh	177 W

At reference site A the unit delivers 0.501 kWh on the solstice and 1.004 kWh on 1 February [B3]. R3 (0.5 kWh on a clear winter day) is **at risk**: site A meets it with no margin on the shortest day, and site B, where the beam strikes the glazing at 58° , misses it. The best result comes from a mirror standing well out from the house on the window's axis (site C, 1.23 kWh on 1 February [B4]), where the house does not shade the mirror and the beam meets the glazing square on, but that site is 15 m away and close to the spill limit in section A. Siting matters more than any hardware choice; the siting survey (HLT-DDR-001, O3) should use this model.

■ C. Mast and yoke stiffness (R4)

The 60.3 x 3.9 mm pipe has $I = 276,079 \text{ mm}^4$ and $EI = 5.52 \times 10^{10} \text{ N}\cdot\text{mm}^2$ [C1]. At 8 m/s face-on the mirror force is 16.9 N, and the mast top tilts 0.040° [C2]. The same load on a 2.2 m 40 x 40 mm extrusion would tilt it 0.37° [C3], which confirms decision D1.

The printed yoke arms carry the wind force and react the hinge moment through the elevation gearbox. With the TRL 2 scaffold's 30 x 50 mm section, 382 mm long, the arm slope at 8 m/s would be 0.145° , three times the 0.05° allowed in the TRL 2 budget. The model therefore uses 40 x 70 mm arms, which give 0.040° [C4]. Mast twist from an assumed yaw moment of 0.85 N·m is 0.002° [C5].

■ D. Pointing error budget and calibration (R4, R6, R7)

A DS3231 at ± 2 ppm drifts 31.5 s in 6 months, during which the sun moves 0.131° ; that is 0.066° as a mirror-normal equivalent [D1]. R6 (60 s) is met. The drive resolves 0.0023° per microstep at the output [D2].

Table 3. Calibration simulation: worst mirror-normal error from 09:00 to 15:00 at site A on 21 December [D3].

Calibration points	Median	95th percentile
3 points over 30 min	9.8°	38.7°
3 points over 3 h	0.18°	0.45°
4 points over 4 h	0.12°	0.23°

Points taken within half an hour cannot separate the five mount errors, and the fit is unusable. Three points over 3 h are adequate on average but leave a long tail. Four points over about 4 h are used below.

Table 4. Pointing error budget, mirror normal [D4, D5].

Source	Normal error	Basis
Sun-position algorithm (Grena 5)	0.001°	Half of 0.0027°
Clock, 6 months without network time	0.066°	[D1]
Mount model residual, 4 points over 4 h	0.116°	Median of [D3]
Worm backlash with preload	0.050°	Assumed lost motion with the teeth held on one flank
Step resolution	0.002°	[D2]
Mast bending, 8 m/s	0.040°	[C2]
Yoke arm flex, 8 m/s	0.040°	[C4]
Root sum square	0.153°	0.31° beam against 0.5°

With the 95th percentile calibration residual (0.23°) the beam error is 0.50° , exactly at the limit [D5]. The budget also depends on a preload: the listed backlash of the gearbox is 1° , and if the wind reverses the load, half of it can appear at the mirror, giving 1.04° beam error [D6]. The mirror's center of mass sits 13.7 mm in front of the elevation axis, a gravity moment of up to $0.80 \text{ N}\cdot\text{m}$ [H4], so a preload of more than $2.9 \text{ N}\cdot\text{m}$ (operating hinge moment $2.1 \text{ N}\cdot\text{m}$ plus imbalance) is needed at 8 m/s [H5]. The decided spiral preload spring of about $3 \text{ N}\cdot\text{m}$ on each axis (HLT-DDR-002 N2, BOM line 16) needs about $0.30 \text{ N}\cdot\text{m}$ at the motor with wind and imbalance [H5], within a 48 mm NEMA17's holding torque but with little margin at speed. The elevation spring biases the mirror toward stow, so it never opposes a stow. R4 remains **at risk**: the typical beam error is 0.31° , but the 95th percentile of calibration sits at the 0.5° limit.

Calibration hands-on time by task analysis is 19 min for four points (5 min setup, 3 min per point, 2 min to fit and save) [G1], and the points span about 4 h of one day [G2]. Four points over about 4 h are now the default (HLT-DDR-002 N6), and R7 now reads "30 min or less of hands-on time, spread over one clear day" (N4), so R7 is **met**.

■ E. Wind, drives, mast and anchor (R9)

Table 5. Wind loads [E1].

Wind	Pose	q	Mirror force	Hinge moment	Mast base moment	Mast stress
8 m/s	Tracking	39 Pa	17 N	2.1 N·m	41 N·m	5 MPa
15 m/s	Tracking (stow trigger)	138 Pa	60 N	7.4 N·m	146 N·m	16 MPa
20 m/s	Tracking	245 Pa	106 N	13.2 N·m	259 N·m	28 MPa
35 m/s	Stowed face-down	750 Pa	162 N	24.3 N·m	436 N·m	48 MPa
35 m/s	Caught face-on	750 Pa	324 N	40.5 N·m	793 N·m	87 MPa

The mast is adequate in every case: caught face-on at 35 m/s it reaches 87 MPa against 240 MPa yield, a factor of 2.8, and the anchor must resist 0.79 kN·m [E4]. The BOM asks for an anchor rated at 1.0 kN·m or more.

The weak link remains the elevation gearbox. With the decided storm awareness (D2), the controller stows when the anemometer sees a 15 m/s gust or the forecast predicts one; the hinge moment at the trigger is 7.4 N·m, within the 17 N·m rating [E5]. In stow, the assumed stowed hinge moment at 35 m/s is 24.3 N·m, above both the 17 N·m rating and the 22 N·m listed maximum [E2]; the rating is reached at a 29 m/s gust [E3]. Without a stow latch R9 would not be met: the gearbox would carry 2.3 N·m more than its listed maximum [E6] (the v0.1 result). On 2026-09-25 Amish decided on stow stops on the yoke that carry the stowed hinge moment in both directions (HLT-DDR-002 N1). A fixed stop can only resist one direction on an axis that must turn into the stow, so the design uses a stop and a latch together:

- A steel lug on the -X end of the torque tube, between the mirror edge and the -X arm, lies along -Y when the mirror is face-down.
- A steel bracket on the -X arm carries a 3 mm polyurethane (90A) pad below the stowed lug, which takes the moment in the stowing direction, and a sprung steel latch pawl above it, which the lug cams past on the way into stow and which then takes the moment in the other direction. The latch engages without power, so a power-loss stow on the supercapacitor reserve also latches.
- A 12 V pull solenoid lifts the pawl for a few seconds when the unit leaves stow; it draws no standby power.
- After latching, the firmware backs the elevation worm off to the middle of its 1° backlash, so the worm teeth do not touch while the pad compresses.

The latch is designed for 48.6 N·m, twice the assumed stowed moment, which covers a stowed hinge moment coefficient up to 0.30 [E7]. This is the check against option C: a published stowed coefficient for small heliostats would have to exceed twice the value assumed here before the latch is overloaded. At that load the contact force is 884 N, the pad stress 4.6 MPa and its deflection 0.28 mm, less than the 0.48 mm the lug can move before the worm teeth touch, and the 8 mm pawl pivot pin sees 9 MPa in double shear [E8]. The gearbox therefore carries no stowed moment, and R9 is **met** on paper. The stowed coefficient and the polyurethane modulus are still assumptions, and the case of a unit caught face-on (40.5 N·m) still exceeds the gearbox, so the stow must not fail.

■ F. Stow on power loss (R10)

A worst-case stow turns the elevation axis 180° at $10^\circ/\text{s}$ in 18 s, plus 5 s to detect the loss, and uses 87 J at 3.8 W [F1]. Five 2.7 V, 25 F supercapacitors in series (5 F) charged to 11.7 V and used down to 7.0 V hold 187 J after 85 % conversion, a margin of 2.1, with 2.34 V per cell against the 2.7 V rating [F2]. The stow always turns the normal downward. From site A at 11:00 on 21 December the beam starts 4.0° below the horizon and never rises above it; the reflection ends after 7.7 s, and for about 2.8 s the beam crosses the ground between the target and a point 3 m from the mast [F4]. The stow finishes in 23 s, within 60 s, and the latch engages without power. R10 now allows the beam to move only downward, toward the ground, during a stow (HLT-DDR-002 N5), so R10 is **met**.

■ G. Calibration time (R7)

See section D and [G1], [G2].

■ H. Mass and balance (R13)

The mirror assembly weighs 5.98 kg: glass 2.70, panel 1.98, torque tube 0.32, ribs 0.50, trunnions 0.29, adhesive and film 0.20 [H1]. The yoke, printed at 60 % infill from the model's 4.23 L, weighs 2.72 kg; each drive 1.66 kg (gearbox 1.2 kg assumed); the turntable 0.5 kg [H2]. The stow stop and latch add 0.27 kg and the two preload springs 0.16 kg [H3]. The total on the mast top is 12.95 kg against the 13 kg limit decided on 2026-09-25 (HLT-DDR-002 N3), a margin of 0.05 kg [H3], so R13 is **met**, thinly and on assumed gearbox, spring and solenoid masses. The TRL 2 estimate of about 8 kg assumed small generic NEMA17 worm gearboxes without a rating and a 0.8 kg yoke; the rated NMRV030-class gearboxes and the stiffer arms of section C add about 4.5 kg. The mast pipe adds 9.2 kg and the cap plate 1.6 kg. Installation in 4 h by two people is not verifiable at TRL 3.

■ I. Power (R14)

Average draw is 0.60 W: idle 0.4 W, moves 0.17 W and 0.029 W in the stow reserve's balancing resistors, about 14 Wh a day [I1]. R14 (3 W) is met.

■ J. Cost (R15)

The BOM has 17 lines, all priced, totaling \$451.00 against the \$455 budget, a \$4.00 margin [J1]. The preload springs (line 16, \$6) and the stow stop and latch (line 17, \$20, of which the solenoid is about \$8) took it \$21 over the earlier \$430 budget; Amish approved a top-up to \$455 on 2026-09-26 (HLT-DDR-002 v0.2). R15 is met. Tools, printer time and shipping are not included [J2].

■ K. Results

The yoke sweeps: at site A the normal stays within 46° of the target direction on the solstice, inside the $\pm 135^\circ$ range; the mirror clears the arms by 20 mm and the crossbar by 46 mm, so a face-down stow clears the yoke; the stow lug runs 3 mm outside the mirror edge and 9 mm inside the -X arm [K1].

Table 6. Requirement status, not met first [K2, K3].

ID	Requirement	Value	Target	Status
R3	Daily energy	0.50 kWh (21 Dec) and 1.00 kWh (1 Feb) at site A; 0.40 kWh at site B on 21 Dec	0.5 kWh or more on a clear winter day	At risk
R4	Pointing	0.31° beam typical, 0.50° at the 95th percentile, with the decided preload springs; 1.04° without preload	0.5° beam up to 8 m/s	At risk
R12	Outdoor life	ASA, galvanized steel, glass with backing film, by selection	-20 to +45 °C, UV, corrosion, 10 years	Not verifiable at TRL 3
R1	Redirected sunlight	211 W at the glazing	200 W or more	Met
R2	Daylight	502 lx mean added (utilization 0.4 assumed)	300 lx or more	Met
R5	Sensorless tracking	Time, location and target only; Hall switches for homing, anemometer for weather	No sun sensor or camera	Met
R6	Timekeeping	32 s after 6 months	60 s or less	Met
R7	Calibration	19 min hands-on for four points over about 4 h of one day	30 min or less of hands-on time, over one clear day; no survey instrument	Met
R8	Working range	Mechanical range met; spot plus error fits a 1.0 m window up to 17 m	3 to 20 m; $\pm 135^\circ$; -90 to +90°	Met
R9	Wind	Stowed moment 24 N·m at 35 m/s carried by the stow stop and latch (designed for 49 N·m), not the gearbox; 7.4 N·m at the stow trigger; mast 87 MPa worst case	Hold R4 to 8 m/s; survive 35 m/s in stow	Met
R10	Safe beam	Stow in 23 s on stored energy (margin 2.1); beam moves only downward; latch engages without power	Target or ground within 3 m when stowed; only downward during a stow; stow within 60 s	Met
R11	Electrical safety	12 V SELV outdoors, IP65 box, listed indoor adapter	SELV, IP65, listed adapter	Met
R13	Mass and install	12.95 kg on the mast top, 0.05 kg margin	13 kg or less; two people, hand tools, 4 h	Met (install time not verifiable at TRL 3)
R14	Standby power	0.60 W average	3 W or less	Met
R15	Cost	\$451, \$4 margin	\$455 or less	Met

Totals: 0 not met, 2 at risk, 1 not verifiable at TRL 3, 12 met [K3].

■ Checks against earlier figures

- **Optical chain.** TRL 2 quoted 245 W intercepted and about 212 W at the glazing; the script gives 244 W and 211 W [A2]. HLT-PRC-001 and the flow diagram now use the script's values.
- **Spill distance.** TRL 2 said spill rises above about 15 m; with 0.5° error and a 1.0 m window the limit is 16.8 m [A6].
- **Daily energy.** TRL 2 quoted about 0.7 kWh from a generic day; the generic day reproduces 0.73 kWh [B1], but the hourly site model gives 0.50 kWh at site A on the solstice [B3]. Documents now quote the site figures.
- **Clock.** TRL 2 quoted about 32 s; the script gives 31.5 s [D1].
- **Pointing.** TRL 2 quoted 0.17° normal and 0.34° beam with a 0.10° calibration residual, 0.10° backlash and 0.05° yoke flex. The new budget is 0.153° normal and 0.31° beam [D5], but only with four-point calibration, preloaded drives and 40 x 70 mm arms.
- **Wind.** The face-on figures match TRL 2 (40 N·m hinge moment; mast stress now 87 MPa including drag on the pipe, up from 78 MPa). Stowed loads were not estimated at TRL 2 and are new.
- **Mass.** TRL 2 quoted about 8 kg on the mast top; now 12.5 kg [H3].
- **Power.** 0.6 W, unchanged [I1].
- **Cost.** TRL 2 quoted \$391 against \$400; v0.1 of this note gave \$425 against the decided \$430; with lines 16 and 17 it is now \$451, against \$455 after the 2026-09-26 top-up [J1].
- **Changes in v0.2 (HLT-DDR-002).** R9 moved from not met (24.3 N·m on the gearbox against 22 N·m maximum) to met (moment carried by the stow latch); R13 moved from not met (12.5 kg against 10 kg) to met (12.95 kg against 13 kg); R7 and R10 moved from at risk to met after rewording; R15 moved from met (\$425) to not met (\$451 against \$430). R3, R4 and R12 are unchanged.
- **Changes in v0.3 (budget top-up).** BUDGET in sizing.py is \$455; R15 moves from not met (\$451 against \$430) to met (\$451 against \$455). No other figure changes.